

COMMONWEALTH OF AUSTRALIA

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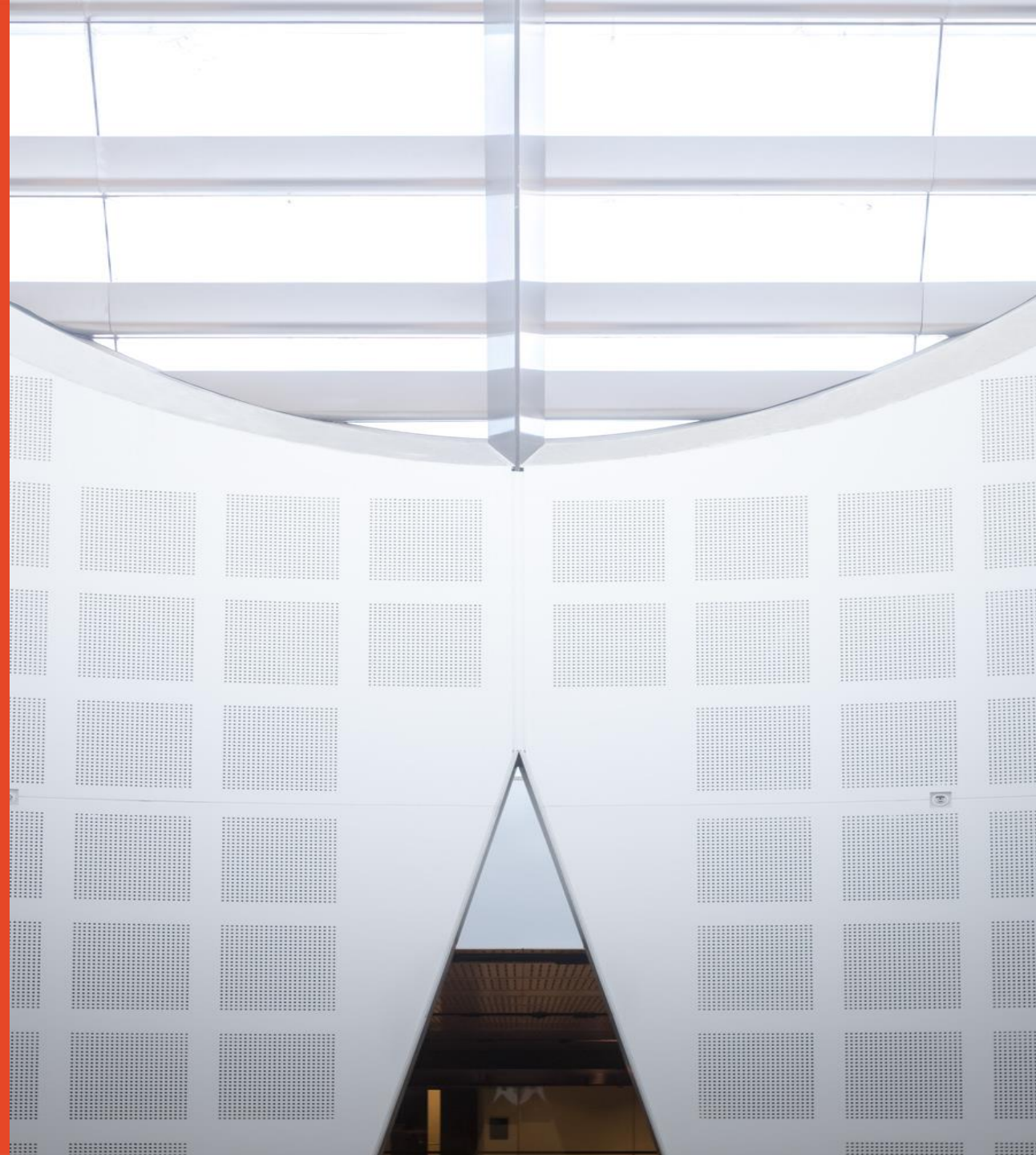
COMPx270: Randomised and Advanced Algorithms

Lecture 3: Balls in Bins

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THE UNIVERSITY OF
SYDNEY



A question 🎂

There are quite a few people in the classroom right now. What are the odds two of you (at least) have the same birthday?

A question

Theorem. (The  paradox) If you gather 23 people in a room, then with probability at least 50% a pair will sharing their birthday.

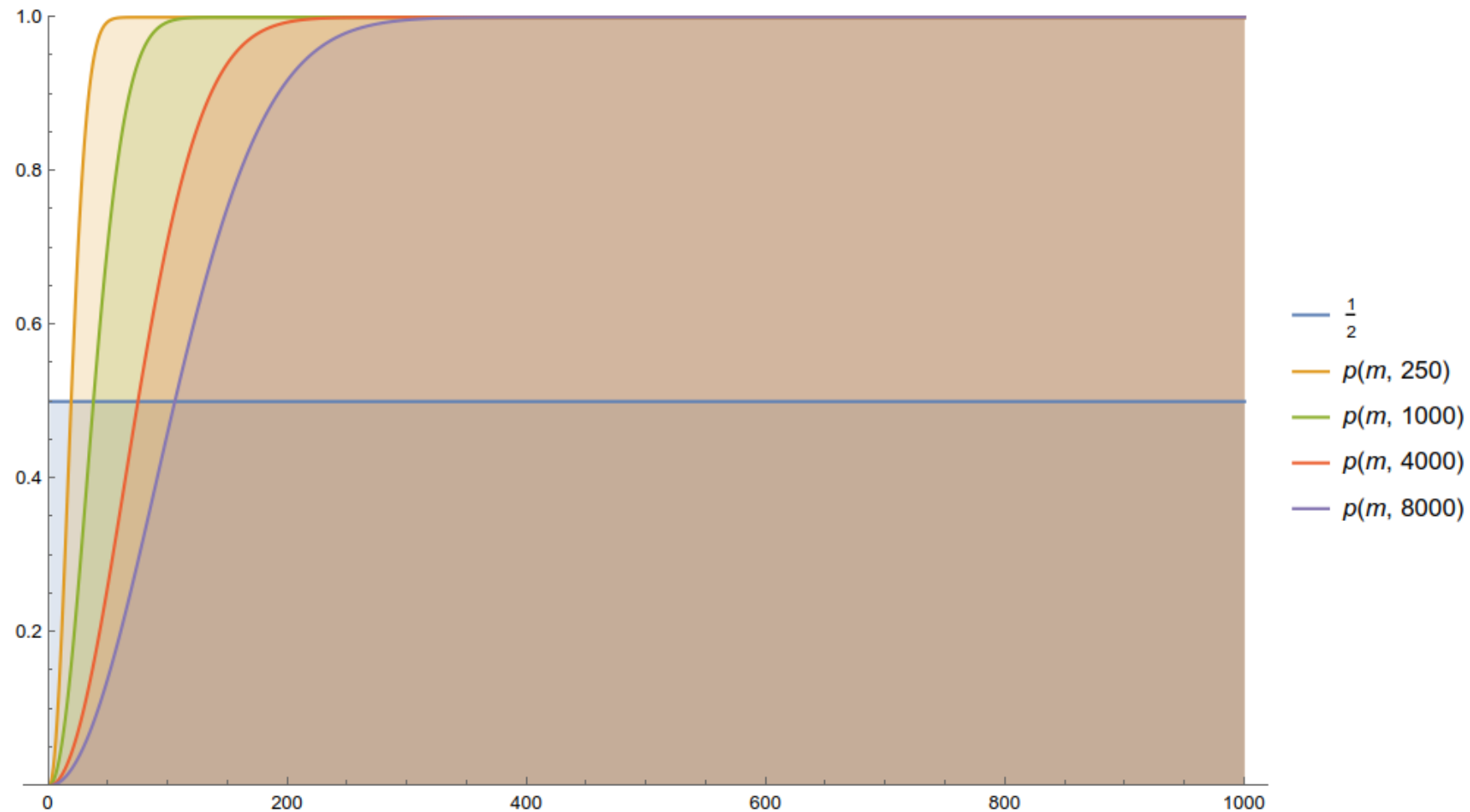
An answer? 🎂

Theorem. If you gather m people and give each a number uniform between 1 and n , then the probability $p(m, n)$ that at least two have the same number is...

$$p_{m,n} = 1 - \frac{n!}{n^m (n - m)!} = 1 - \frac{m!}{n^m} \binom{n}{m}$$

Proof.

```
DiscretePlot[{1/2, p[m, 250], p[m, 1000], p[m, 4000], p[m, 8000]}, {m, 1, 1000}, PlotRange -> {0, 1}, PlotLegends -> "Expressions"]
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Let's start simple: $m=2$  

$X =$ bin first ball falls in

$Y =$ second

$$\Pr[X=Y] = \sum_{i=1}^n \Pr[X=i, Y=i] \stackrel{\text{indep.}}{\approx} \sum_{i=1}^n \underbrace{\Pr[X=i]}_{\frac{1}{n}} \underbrace{\Pr[Y=i]}_{\frac{1}{n}} = \sum_{i=1}^n \frac{1}{n^2} = \frac{1}{n}$$

Now, for large values of "2"...

C: number of collisions when throwing m  into n . What is

$$c(m, n) = E[C]?$$

I have $\binom{m}{2}$ possible collisions (each pair of balls could collide)

$$C = \sum_{(i,j) \text{ pair}} \mathbb{1}_{\text{balls } i,j \text{ collide}}$$
$$E[C] = \sum_{(i,j) \text{ pair}} E[\mathbb{1}_{\text{balls } i,j \text{ collide}}] = \sum_{(i,j) \text{ pair}} \Pr[\text{balls } i,j \text{ collide}] \stackrel{\text{linearity of } E}{=} \binom{m}{2} \frac{1}{n} (= \Theta(\frac{m^2}{n}))$$



Number of collisions

... and what is $\text{Var}[C]$?

$$\text{Var } C = \text{Var} \left[\sum_{(i,j) \text{ pair}} \mathbb{1}_{i,j \text{ collide}} \right]$$

If $\mathbb{1}_{i,j}$ s were independent, this would be $C \sim \text{Bin}(\binom{m}{2}, \frac{1}{n})$, $\text{Var } C = \binom{m}{2} \frac{1}{n} (1 - \frac{1}{n})$
They are not independent!

so...

$$\text{Var } C = \underset{\substack{\uparrow \\ \text{not from}}}{\mathbb{E}[C^2]} - \underset{\substack{\uparrow \\ \text{we know!}}}{\mathbb{E}[C]^2}$$

Number of collisions

... and what is $\text{Var}[C]$?

$$\begin{aligned}\mathbb{E}[C^2] &= \mathbb{E}\left[\left(\sum_{e=(i,j)} \underbrace{1}_{\substack{\text{collide} \\ Y_e}}\right)^2\right] = \mathbb{E}\left[\left(\sum_{e=1}^{\binom{m}{2}} Y_e\right)^2\right] \\ &= \mathbb{E}\left[\sum_e \sum_{e'} Y_e Y_{e'}\right] \\ &\stackrel{\text{linearity}}{=} \sum_e \sum_{e'} \mathbb{E}[Y_e Y_{e'}] \\ &\quad (\text{can be done}) \quad \binom{m}{2} \frac{1}{n} \left(1 - \frac{1}{n}\right) + \binom{m}{2}^2 \frac{1}{n^2}\end{aligned}$$

$$\begin{aligned}\dots \quad \text{so } \text{Var}[C] &= \mathbb{E}[C^2] - \mathbb{E}[C]^2 \\ &= \binom{m}{2} \frac{1}{n} \left(1 - \frac{1}{n}\right) \quad \checkmark\end{aligned}$$



Number of collisions

$$\text{Var}[C] = \binom{m}{2} \frac{1}{n} \left(1 - \frac{1}{n} \right)$$

Number of collisions

$$\text{Var}[C] = \binom{m}{2} \frac{1}{n} \left(1 - \frac{1}{n} \right) \\ \leq \frac{m^2}{2n}$$

Now, we can use Chebyshev:

$$\Pr[X = 0] \leq 1/2$$

for $m = \Omega(\sqrt{n})$.

Number of collisions

$$\text{Var}[C] = \binom{m}{2} \frac{1}{n} \left(1 - \frac{1}{n} \right)$$

Now, we can use Chebyshev:

$$\Pr[X = 0] \leq 1/2$$

for $m = \Omega(\sqrt{n})$. **Is it tight?**

Number of collisions

$$\text{Var}[C] = \binom{m}{2} \frac{1}{n} \left(1 - \frac{1}{n} \right)$$

Now, we can use Chebyshev:

$$\Pr[X = 0] \leq 1/2$$

for $m = \Omega(\sqrt{n})$.

By Markov, we also have

$$\Pr[X \neq 0] = \Pr[X \geq 1] \leq E[X] \leq 1/2$$

for $m = O(\sqrt{n})$.

Applications?

Many!

Bounding the variance: is it always that bad?

Two tricks (and even 3).

① $\text{Var}(\sum_i X_i) \neq \sum_i \text{Var} X_i$ in general
but "often" $\text{Var}(\sum_i X_i) \leq \sum_i \text{Var}[X_i]$
(when $X_1, \dots, X_n$ are negatively correlated)

② $\text{Var}[X] \leq \mathbb{E}[X^2]$ always.

③ "Pairwise independence is enough" for $\text{Var} \sum = \sum \text{Var}$. (We'll get back to it.)

Coverage (Coupon collector)



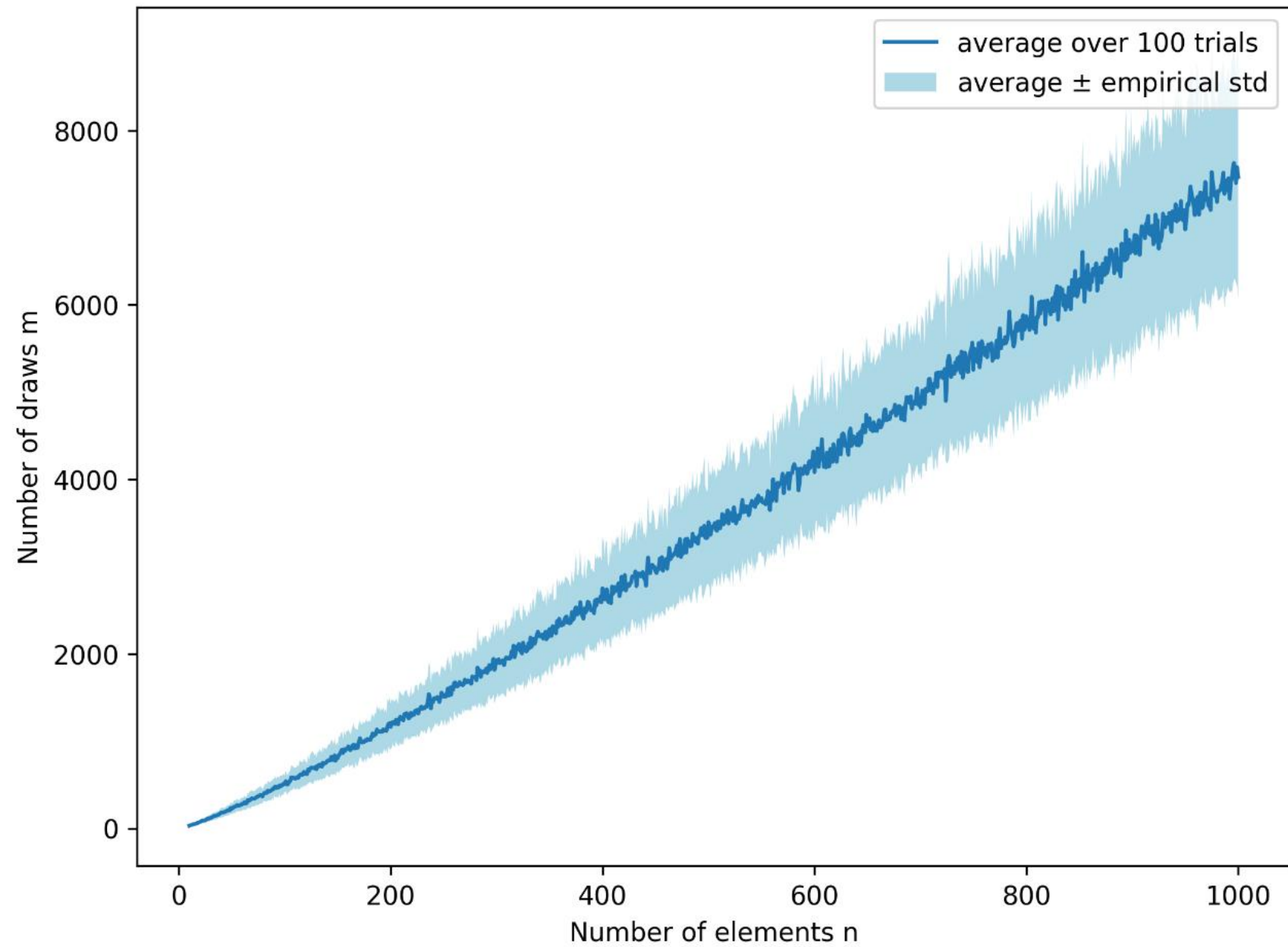
"What is the expected number of balls $M(n)$ we need to throw before each of the n bins contains **at least one** ball?"

Coverage (Coupon collector)



"What is the expected number of balls $M(n)$ we need to throw before each of the n bins contains **at least one** ball?"

- $\Theta(n)$?
- $\Theta(n \log n)$?
- $\Theta(n^2)$?
- Something else?



Coverage (Coupon collector)



"What is the expected number of balls $M(n)$ we need to throw before each of the n bins contains **at least one** ball?"

Theorem. In expectation, $M(n) = \Theta(n \log n)$ balls. (Even more precisely: $n \ln n + O(n)$.)

Intuition: throw n balls.

$$\Pr[\text{no Pikachu}] = \left(1 - \frac{1}{n}\right)^n \approx \frac{1}{e}$$

so $\Pr[\exists \text{ some Pokemon not seen}] \leq n \cdot \left(1 - \frac{1}{n}\right)^n \approx \frac{n}{e}$

$$\mathbb{E}[\# \text{ Pokemon not seen}] \stackrel{\text{union bound}}{\leq} n \left(1 - \frac{1}{n}\right)^n \approx \frac{n}{e}$$

Repeat: $\frac{n}{e^2}, \frac{n}{e^3}, \dots$ until $\ll 1$ Pokemon not seen (in expectation)
 $\rightarrow O(\log n)$ repetitions are enough.

Proof.

T_i = conditioned on having caught $i-1$ distinct Pokémon, what is the # of Pokémon I need to catch before seeing a new one

$$T_1 = 1 \quad \text{and}$$

$$M(n) = T_1 + T_2 + \dots + T_n$$

$$\mathbb{E}[M(n)] = \sum_{i=1}^n \mathbb{E}[T_i] = \sum_{i=1}^n \frac{n}{n-i+1} = n \sum_{i=1}^n \frac{1}{n-i+1}$$

$$= n \sum_{j=1}^n \frac{1}{j}$$

$$= n \cdot H_n$$

$$= n(\ln n + o(1))$$

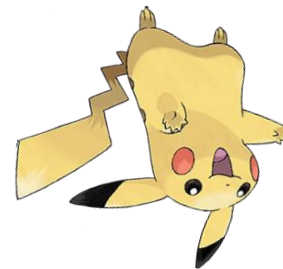
□

Fix i .

$\mathbb{E}[T_i]$?

$$\Pr[\text{getting } i^{\text{th}} \text{ new} \mid \text{got } i-1 \text{ distinct}] = \frac{n-(i-1)}{n}$$

$$T_i \sim \text{Geometric}\left(\frac{n-i+1}{n}\right)$$



What about the variance?



$$\text{Var}(M(n)) = \text{Var}(T_1 + T_2 + \dots + T_n)$$

independent
("magic")

$$\sum_{i=1}^n \text{Var } T_i$$

Variance of Geometric

$$\rightarrow = \sum_{i=1}^n \frac{n^2}{(n-i+1)^2} \underbrace{\left(1 - \frac{n-i+1}{n}\right)}_{\leq 1}$$

$$\leq \sum_{i=1}^n \frac{n^2}{(n-i+1)^2} = n^2 \sum_{j=1}^n \frac{1}{j^2}$$

$$\leq n^2 \sum_{j=1}^{\infty} \frac{1}{j^2} = \boxed{\frac{\pi^2}{6} n^2}$$

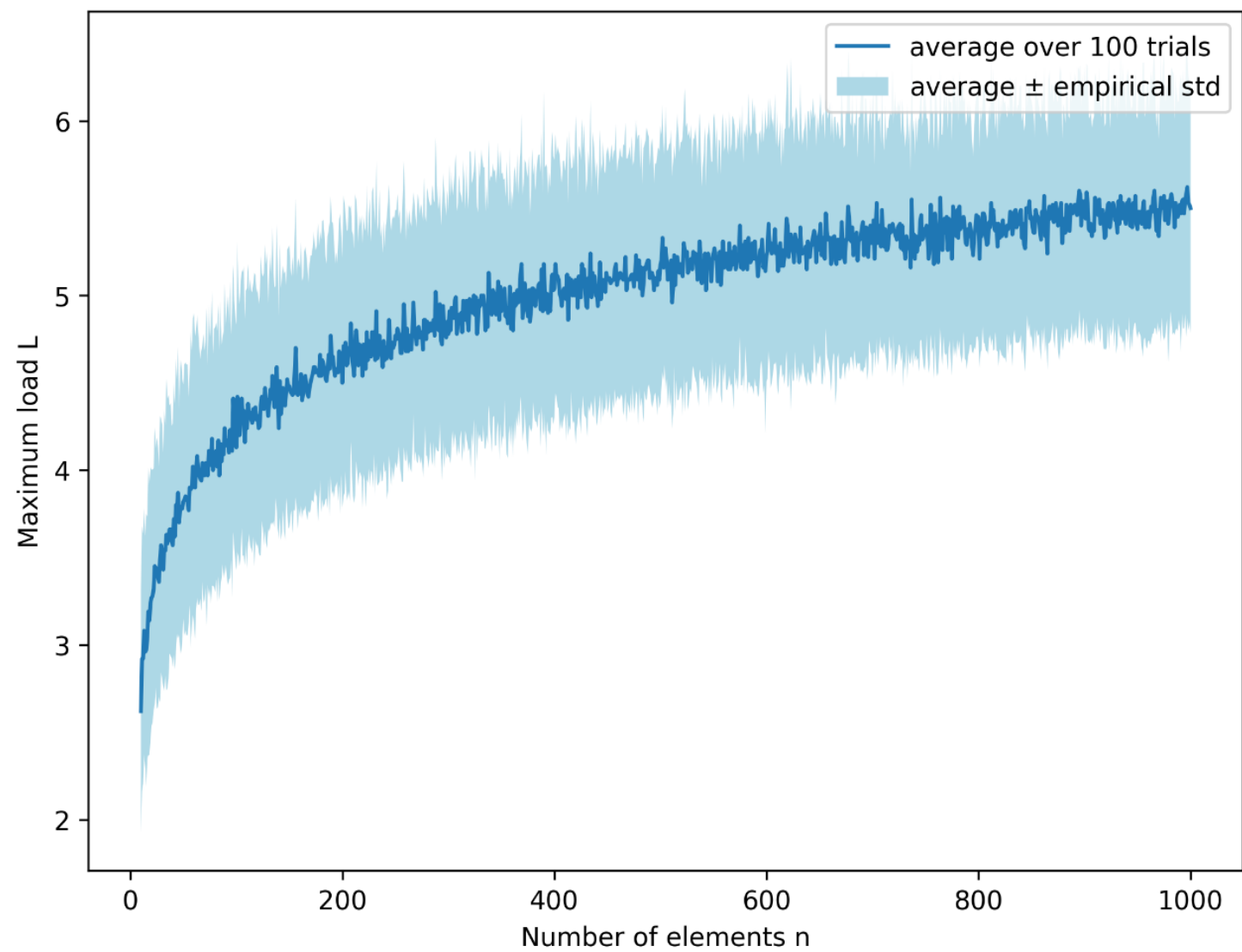
Load balancing

"What is the expected number of balls $L(n)$ the fullest of the n bins contains after throwing n balls?"

Load balancing

"What is the expected number of balls $L(n)$ the fullest of the n bins contains after throwing n balls?"

- $\Theta(1)$?
- $\Theta(\log n)$?
- $\Theta(\sqrt{n})$?
- Something else?



Load balancing

"What is the expected number of balls $L(n)$ the fullest of the n bins contains after throwing n balls?"

Theorem. The expected maximum load is $L(n) = \Theta(\log n / \log \log n)$.

$$\begin{aligned} E[L] &= E\left[\max_{1 \leq i \leq n} L_i\right] = \sum_{k=1}^n \underbrace{P_r[L \geq k]}_{\text{union bound}} \leq n \cdot \sum_{k=1}^n P_r[L_1 \geq k] \\ P_r[L_1 \geq k] &\leq \sum_{i=1}^n P_r[L_i \geq k] \\ &= n P_r[L_1 \geq k] \quad \star \end{aligned}$$

Proof.

Proof.

Fix k $\Pr[L \geq k] \leq \binom{n}{k} \cdot \left(\frac{1}{n}\right)^k \stackrel{?!?}{=} \frac{n!}{k!(n-k)!} n^k$

↑ upper bound to be useful

↑ not useful (hard to manipulate)

Proof.

$$\Pr[L_1 \geq k] \leq \binom{n}{k} \frac{1}{n^k} \leq \frac{e^k n^k}{k^k n^k} = \frac{e^k}{k^k} \quad \checkmark$$

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} \leq \left(\frac{e \cdot n}{k}\right)^k$$

$$\text{so } \mathbb{E}[L] \leq n \sum_{k=1}^n \Pr[L_1 \geq k] \leq \sum_{k=1}^n \frac{ne^k}{k^k}$$

$$\text{We write } \Pr[L \geq k] \leq n \Pr[L_1 \geq k] \leq \frac{ne^k}{k^k}$$

instead:

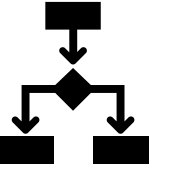
$$\begin{aligned} \mathbb{E}[L] &\leq \sum_{k=1}^n \Pr[L \geq k] \leq \sum_{k=1}^n \min\left(1, \frac{ne^k}{k^k}\right) \\ &\leq \underbrace{\sum_{k=1}^l 1}_l + \sum_{k=l+1}^n \frac{ne^k}{k^k} \end{aligned}$$

≥ 1 for small k

for l to choose

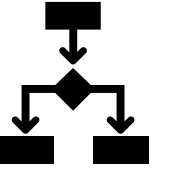
$$\text{for } l = O\left(\frac{\log n}{\log \log n}\right), \text{ get } \mathbb{E}[L] = O\left(\frac{\log n}{\log \log n}\right)$$

Load balancing (a twist)



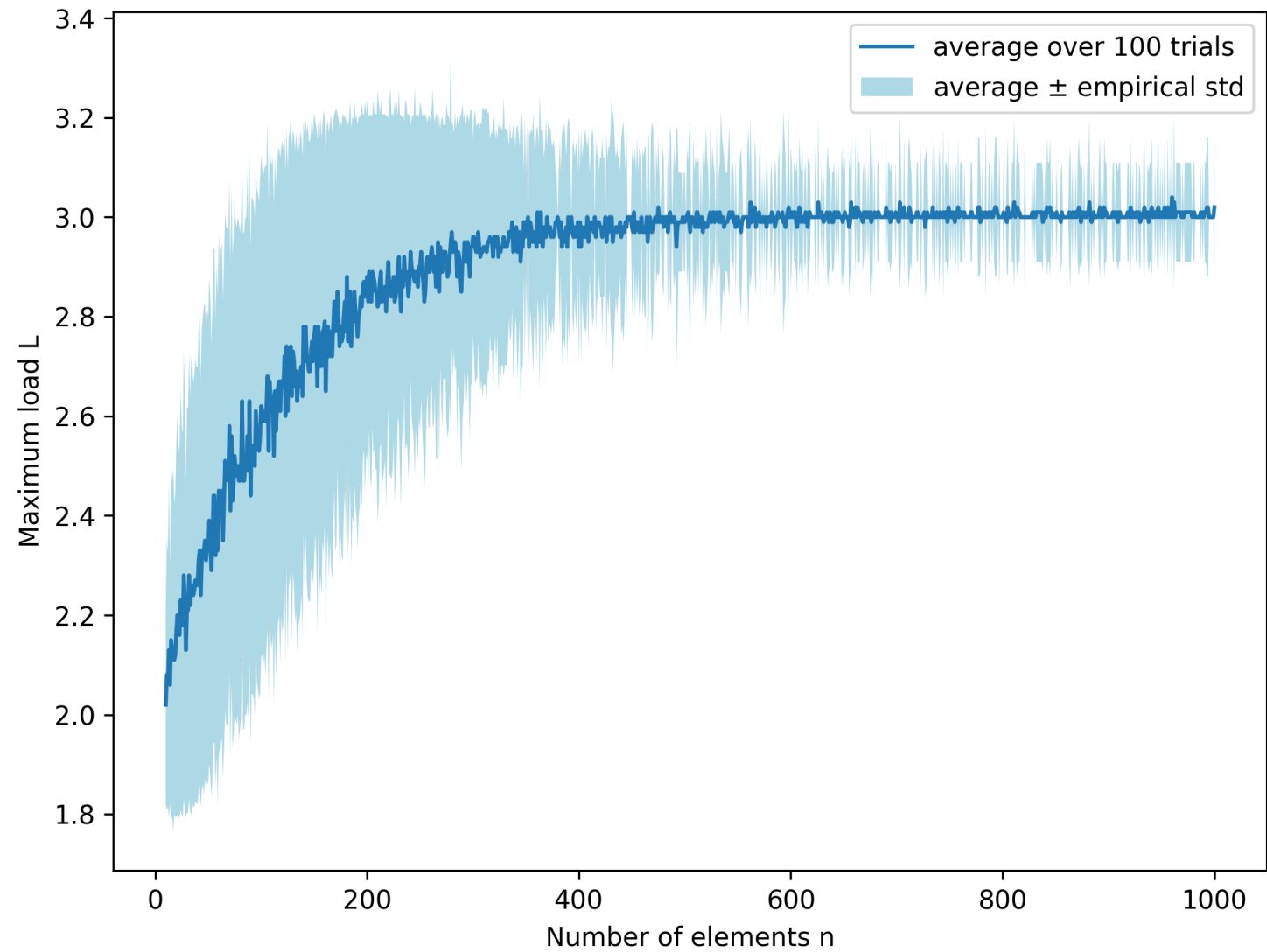
"Now, every time you throw a ball, it selects **two** bins at random, and goes to the least full of the two. What is the maximum expected load?"

Load balancing (a twist)

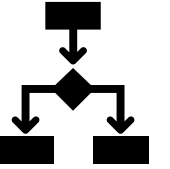


"Now, every time you throw a ball, it selects **two** bins at random, and goes to the least full of the two. What is the maximum expected load?"

- $\Theta(1)$?
- $\Theta(\log \log \mathbf{n})$?
- $\Theta(\sqrt{\log \mathbf{n}})$?
- Something else?

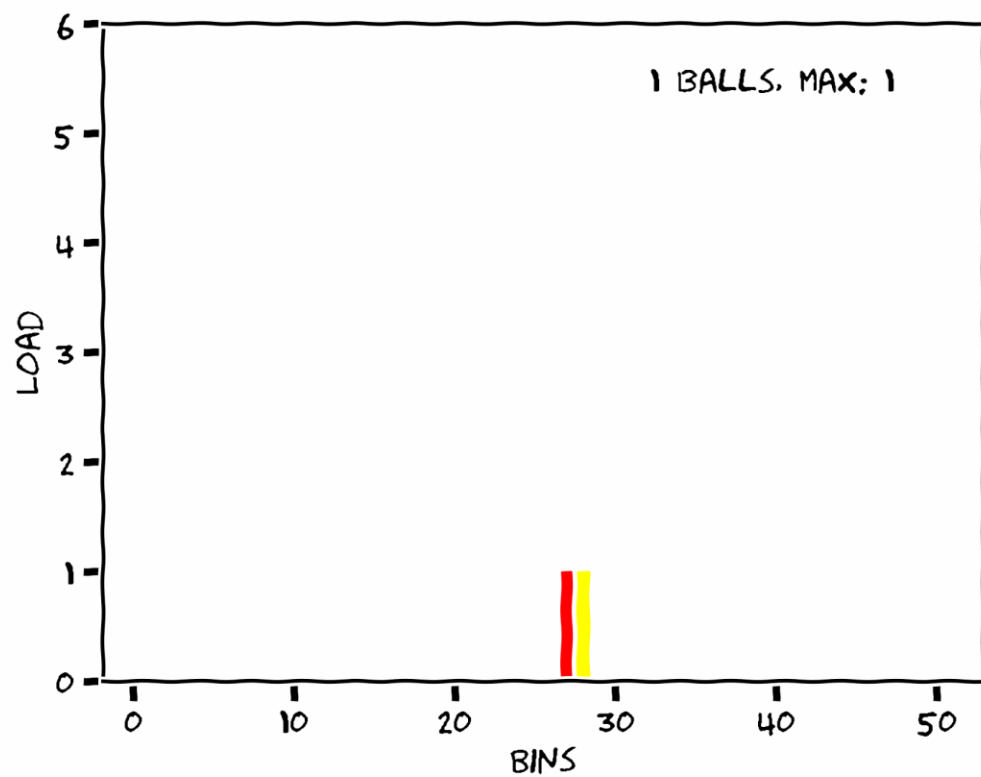


Load balancing (a twist)

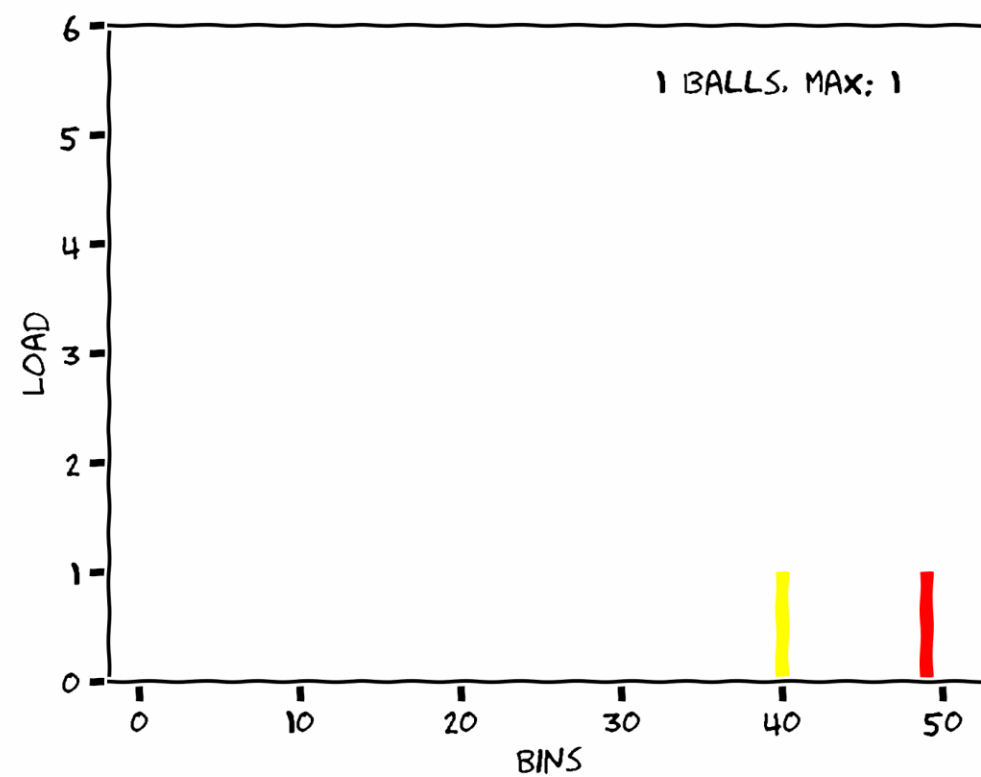


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Theorem. The expected maximum load now $\Theta(\log \log n)$.



"Standard" Balls in Bins



Power of Two Choices

Conclusion

